

Sri Nayan Reddy Sama

Kalamazoo, MI | P: +1 2693641372 | samanayan05@gmail.com

SUMMARY

I am a graduate student in ECE with experience in system modeling, control simulations, and automation using MATLAB/Simulink and Python.

EDUCATION

M.S. Electrical & Computer Engineering, Western Michigan University (Aug 2023 – Jun 2025) | GPA: 3.05
B. Tech ECE, MLR Institute of Technology (Aug 2019 – May 2023) | GPA: 3.5

RELEVANT WORK EXPERIENCE

GRADING ASSISTANT Western Michigan University

Kalamazoo, MI

ECE3800 – Probability of system and signal analysis

August 2023 – December 2023

- As a Grading Assistant for ECE3800 – **Probability of System and Signal Analysis**, I meticulously evaluated assignments, quizzes, and exams for over 25 students, ensuring fair and accurate grading. I played a pivotal role in curating course materials, including complex problem sets and lecture notes, to enhance student comprehension. My guidance significantly bolstered student performance, leading to a notable improvement in the overall class average by at least 30%.

ECE3610 – Electromagnetic Fields

August 2024- December 2024

- For the Fall 2024 **Electromagnetic Fields** course, I evaluated 20 students on concepts including static and time-varying electric and magnetic fields, plane and guided waves, transmission lines, radiation, and antennas. My responsibilities included grading assignments, exams, and ensuring academic integrity by checking for plagiarism.

PART-TIME RESEARCH ASSISTANT – Dr. Ikhlas Abdel Qader

Kalamazoo, MI

ECE6970

January 2024 – April 2024

- Designed a **high-precision object detection system for autonomous vehicles** by integrating Lidar, Camera, Radar, and Ultrasonic sensors to improve detection accuracy in adverse weather conditions. **Implemented the YOLOv8 algorithm in Python for application configuration**, using the latest COCO dataset of 15,000 images and 14,000 annotations. **Comparative analysis** with previous models showed enhanced detection performance, setting a strong foundation for future real-time, multi-modal sensor fusion applications.

RELEVANT PROJECTS

- **Reinforcement Learning for Fault Isolation in Inverter-Based Microgrids** April 2025-June 2025
Developed a simulation model of a microgrid with PV, inverter, load, and fault scenarios using **MATLAB/Simulink**. Implemented **common fault types** (3-phase, L-G) and monitored system responses via voltage, current, and frequency sensors. Designed and proposed a **Reinforcement Learning (RL) agent** to optimize fault response actions (breaker toggling, load shifting). Explored **Q-Learning and DDPG** methods for training an agent on simulated data exported from Simulink. Analyzed RL performance metrics in terms of **fault isolation speed, voltage recovery, and reward efficiency**.
- **Designing Powertrain of an EV using MATLAB Simulink** January 2024 – March 2024
Engineered an advanced powertrain model for the Koenigsegg Gemera hyper car in MATLAB Simulink, incorporating detailed subsystems for electric motors, battery management, and drive ratios. Employed highway and city drive cycle datasets to simulate and optimize vehicle range, efficiency, and performance parameters. Achieved precise outcomes in battery depth of discharge, state of charge, acceleration profiles, power output, and current draw during dynamic vehicle motion. Used simulated sensor feedback and power output testing to optimize dynamic performance; verified results with scope-like signal traces.

SKILLS

Programming: Python, MATLAB/Simulink, C | **Tools:** LabVIEW, PLC Basics, SCADA, HIL, SIL, CAN protocols | **Simulation:** ANSYS, LTSpice | **Computer Vision:** YOLO-V8, Image Segmentation